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Chemical Bonding and Hybridisation — Issue #32 specimen: Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

In the introductory valence-bond model,
name the hybrid-orbital set used for four

Representation

equivalent electron-domain directions around a central atom, and state how many hybrid orbitals it contains.

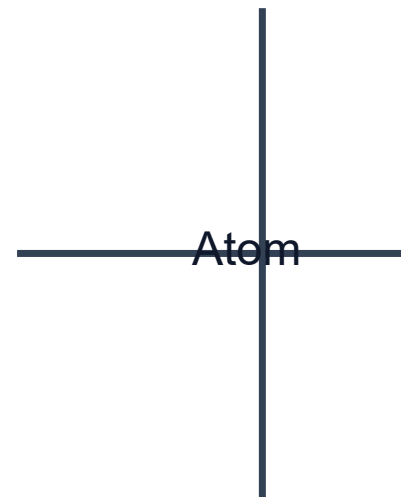
Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

+

Four supplied equivalent directions



Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

+

Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Steric electron-domain counting and multiple bonds

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric electron-domain counting and multiple-bond constraints

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The Lewis structure $\text{H}_2\text{C}=\text{O}$ has a double bond from carbon to oxygen and one single bond from carbon to each hydrogen. Explain why the $\text{C}=\text{O}$ double bond counts as one electron domain around carbon.

Representation

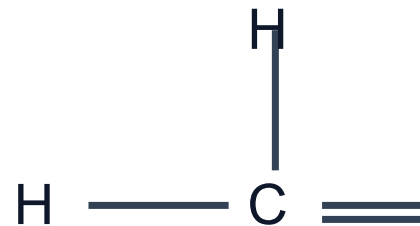
Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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H₂C=O supplied Lewis connectivity



H₂C=O connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Steric electron-domain counting and multiple bonds

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For CH₄, use the known rule for four bonding domains and no central lone pair to assign the carbon hybridisation and electron-domain geometry. Show the domain count.

Representation

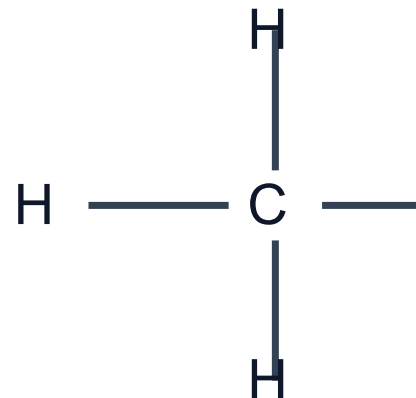
Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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CH₄ supplied connectivity



CH₄ connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: 3D Stereochemical representation and qualitative models

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

NH₃ has three N–H single bonds and one lone pair on nitrogen. Translate that information into a labelled diagram showing all four electron domains; distinguish the lone pair from the three

Representation

bond directions. Do not treat the planar Lewis sketch as the molecular shape.

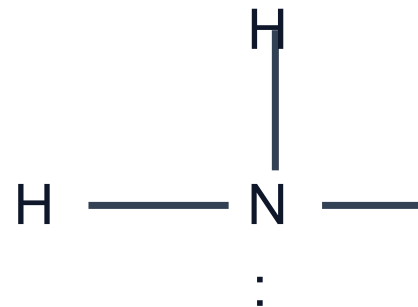
Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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NH₃ supplied Lewis inventory



NH₃ Lewis inventory

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

3D Stereochemical representation and qualitative models

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: 3D Stereochemical representation and qualitative models

Concept difficulty: Medium concept

Question difficulty: Multi-step reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A model is needed only to describe the overall three-dimensional arrangement of electron domains in NH_3 ; it need not describe the detailed orbital overlap. Choose between VSEPR and a hybrid-orbital overlap diagram, and identify what

Representation

your chosen model predicts and what it leaves unresolved.

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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Requested output: domain arrangement

Candidate: VSEPR

Candidate:

Overall domain arrangement requested

Detailed overlap excluded

Connectivity layout only; page angles are not measured given
Supplied situation

Need a hint? · guided support

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Need the concept again?

3D Stereochemical representation and qualitative models

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For a qualitative orbital sketch, overlap A is end-on along the internuclear axis and overlap B is side-on on opposite sides of that axis. Classify each as sigma or pi overlap and identify the defining feature you used.

Representation

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning



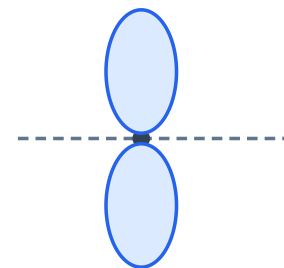
Supplied overlap situations A and B

A: end-on



Same nucleus-to-nucleus reference

B: side-on



Same nucleus-to-

Connectivity layout only; page angles are not measured geometry.

Supplied situation

Need a hint? · guided support



Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: 3D Stereochemical representation and qualitative models

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For CH₄, sketch a three-dimensional arrangement of the four C–H bond directions using a wedge/dash convention or another clearly defined convention. Explain how your drawing differs from a flat cross-shaped Lewis sketch.

Conditions

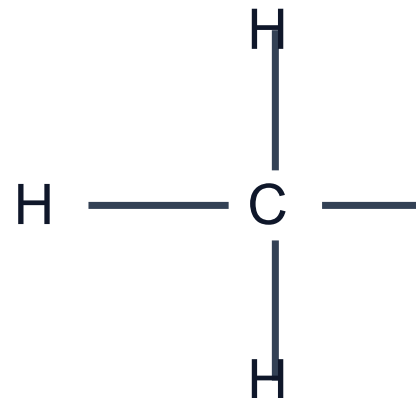
- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Representation

**Common wrong route · open if you
want a warning**

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CH₄ connectivity to be translated



CH₄ connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

3D Stereochemical representation and qualitative models

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Steric electron-domain counting and multiple bonds

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric electron-domain counting and multiple-bond constraints

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A nitrogen atom in NF_3 has three N–F single bonds and one lone pair. Each fluorine also has lone pairs. Count the electron domains around nitrogen and state which lone pairs belong in that central-atom count.

Representation

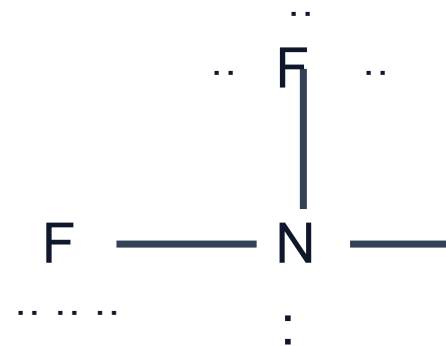
Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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NF3 atom-local inventory



NF3 Lewis inventory

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Steric electron-domain counting and multiple bonds

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Steric electron-domain counting and multiple bonds

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric electron-domain counting and multiple-bond constraints

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A student counts the double bond in $\text{H}_2\text{C}=\text{O}$ as two separate directions and predicts four domains around carbon. Locate the counting error and give the corrected domain count.

Representation

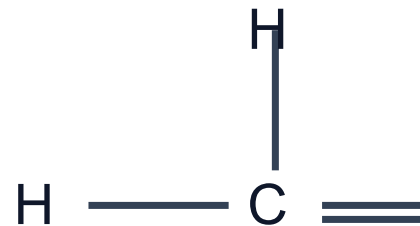
Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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H₂C=O counting operation to audit



H₂C=O connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Steric electron-domain counting and multiple bonds

Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

The benchmark supplies this table: BF₃ has three bonding domains and no central lone pair; CH₄ has four bonding domains and no central lone pair. Match each central atom to the introductory sp² or sp³ model, and check that the number of hybrid-

Representation

orbital directions agrees with the domain count.

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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Supplied BF₃ and CH₄ inventories



BF₃ Lewis inventory

CH₄ con

Connectivity layout only; page angles are not measured ge
Supplied situation

Need a hint? · guided support

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Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

Chemistry · Chemical Bonding and Hybridisation — Issue #32 specimen